Dr. Seung-Goo Kim
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MAX PLANCK INSTITUTE
FOR EMPIRICAL AESTHETICS

# Curriculum Vitae

## Personal Data

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|---|---|
| Title | Dr. rer. nat. (Ph.D.) |
| First name | Seung-Goo |
| Name | Kim |
| Current position | Research Scientist (until 2026.08.31) |
| Current institution(s) / site(s), country | Max Planck Institute for Empirical Aesthetics, Frankfurt am Main, Germany |
| Identifiers/ORCID | 0000-0003-0558-8547 |

## Qualifications and Career

| **Stages** | **Periods and Details** |
|---|---|
| High school | **Musical composition**, March 1996 – February 1999, Sunwha Art School, Seoul, South Korea |
| Bachelor's program | **Economics** and **psychology** (dual majors), March 2000 –February 2004, Yonsei University, Seoul, South Korea |
| Master's program | **Cognitive science**, September 2008 – July 2010, Seoul National University, Seoul South Korea<br>- Advisor: Dr. Chun Kee Chung<br>- Thesis: "The effect of conditional probability of chord progression in Western music corpus on brain response: an MEG study" |
| Doctorate | **Psychology**, October 2012 – January 2017, International Max Planck Research School on Neuroscience of Communication, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany<br>- Advisors: Dr. Thomas R. Knösche (MPI-CBS) and Dr. Erich Schröger (University of Leipzig)<br>- Dissertation: "Myeloarchitecture and Intrinsic Functional Connectivity of Auditory Cortex in Musicians with Absolute Pitch" |
| Stages of academic / professional career | **Postdoctoral research associate**, October 2017 – August 2018, Department of Psychiatry, Behavioural and Clinical Neuroscience Institute, University of Cambridge, Cambridge, UK<br>- Supervisor: Dr. Valerie Voon<br>- Projects: intracortical myelination and its association with impulsivity and alcohol consumption<br><br>**Postdoctoral research associate**, September 2018 – August 2021, Department of Psychology and Neuroscience, Duke University, Durham, NC, USA<br>- Supervisor: Dr. Tobias Overath<br>- Projects: MEG correlates of auditory percepts, fMRI encoding of natural speech and music |

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|---|---|
| | **Postdoctoral researcher**, September 2021 – August 2026,<br>Research Group Neurocognition of Music and Language, Max Plank Institute for Empirical Aesthetics, Frankfurt am Main, Germany<br>- Supervisor: Dr. Daniela Sammler<br>- Projects: fMRI encoding of musical emotions |

## Supplementary Career Information

Served mandatory military duty as an officer in Republic of Korea Air Force, honourably discharged as a first lieutenant (from April 2004 to July 2007).

## Engagement in the Research System

*Guest editor*

2023–Present Proceedings of the National Academy of Sciences of the United States of America (PNAS)

*Review editor*

2022–Present Frontiers in Neuroscience, specialty section: Auditory Cognitive Neuroscience

*Ad-hoc reviewer for journals (in an alphabetical order)*

2014–Present Brain and Behavior (2), Brain Connectivity (3), Brain Research (6), Brain Structure and Function (1), Cognitive Neurodynamics (2), eLife (1), Frontiers in Aging Neuroscience (3), Frontiers in Computational Neuroscience (1), Frontiers in Human Neuroscience (1), Frontiers in Neuroscience (2), Frontiers in Psychology (2), Human Brain Mapping (1), International Journal of Psychophysiology (6), Journal of Integrative Neuroscience (1), Journal of Physiological Anthropology (1), NeuroImage (5), Neuropsychologia (1), Neuroscience Letters (1), PLOS ONE (2), Psychophysiology (2), Scientific Reports (2)

*Academic Organization Member*

2023-Present International Conference on Music Perception and Cognition (https://icmpc.org/)- Abstract reviewer (2023–Present), Session chair (2023–Present)

2012–Present Organization for Human Brain Mapping (https://www.humanbrainmapping.org/)- Abstract reviewer (2021–Present)

2011–Present Society for Neuroscience (https://www.sfn.org/)

*Teaching*

2022–present "Methods in Empirical Musicology", a practical course (seminar and hands-on) for undergraduate and graduate students majoring in Music Informatics, Karlsruhe University of Music (Hochschule für Musik), Karlsruhe, Germany (online).

## Supervision of Researchers in Early Career Phases

2023 Katarzyna Gugnowska. (Ph.D. student at NCML, MPI for Empirical Aesthetics, Frankfurt; PI: Daniela Sammler)
- Project: Neural encoding of piano duet
- Tasks: EEG data analysis using the temporal response function modeling

2022 Zoë Bolz. (undergraduate intern at NCML, MPI for Empirical Aesthetics, Frankfurt; PI: Daniela Sammler)
- Project: Linguistic knowledge and cultural exposure on prosody perception
- Tasks: behavioral data analysis with R

2021–2022 Han (Medy) Mu. (undergraduate researcher at O-lab, Duke University, NC; PI: Tobias Overath)

- Project: Neural encoding of music in fMRI data
- Tasks: MATLAB & Bash for fMRI data processing and linearized encoding analysis

2021–2022 Ying Yu. (undergraduate researcher at O-lab, Duke University, NC; PI: Tobias Overath)
- Project: Temporal structures in tonal and atonal music: online behavioral experiment
- Tasks: MATLAB, JavaScript, HTML for MTurk experiment & analysis

2019–2021 Jasmine Leahy. (undergraduate researcher at O-lab, Duke University, NC; PI: Tobias Overath)- Project: Neural encoding of musical features in EEG data
- Tasks: MATLAB scripting for EEG experiments and data analysis
- B.Sc. thesis (graduation with distinction): "Cortical Encoding of Tonal and Rhythmic Hierarchy in Real and Imagined Music Using the Multivariate Temporal Response Function"
- Current position: M.D. student at the Icahn School of Medicine at Mount Sinai, NY

2018–2020 Jie Wan. (graduate researcher at O-lab, Duke University, NC; PI: Tobias Overath)
- Project: Cortical entrainment to temporal structures in natural speech
- Tasks: MATLAB scripting, EEG data analysis, encoding analysis (mTRF), statistical tests, data visualization
- Current position: Ph.D. student at University of California – Irvine, CA

## Scientific Results

*Journal papers on auditory perception (N = 12)*

2024 Kim S-G, De Martino F, Overath T. (accepted). Linguistic modulation of the neural encoding of phonemes. *Cerebral Cortex*.

2022 Kim S-G. (2022, Sep). On the encoding of natural music in computational models and human brains. *Frontiers in Neuroscience 16: 928841*. doi:10.3389/fnins.2022.928841

Kim S-G*, Overath T*, Sedley W, Kumar S, Teki S, Kikuchi Y, Patterson R, Griffths TD. (2022, Jan). MEG correlates of temporal regularity relevant to pitch perception in human auditory cortex. *NeuroImage 249: 118879*. doi:10.1016/j.neuroimage.2022.118879 (*Both share first authorship).

2021 Leahy J*+, Kim S-G*, Wan J, Overath T. (2021, Jul). An Analytical Framework of Tonal and Rhythmic Hierarchy in Natural Music Using the Multivariate Temporal Response Function. *Frontiers in Neuroscience 15: 665767*. doi:10.3389/fnins.2021.665767. (*Both share first authorship; +Undergraduate first-author)

2020 Kim S-G, Poeppel D, Overath T. (2020, Feb). Modulation change detection in human auditory cortex: Evidence for asymmetric, nonlinear edge detection. *European Journal of Neuroscience 52(2): 2889-2904*. doi:10.1111/ejn.14707

2019 Kim S-G, Mueller K., Lepsien J, Mildner T, Fritz TH. (2019, Dec). Brain networks underlying aesthetic appreciation as modulated by interaction of the spectral and temporal organisations of music. *Scientific Reports 9: 19446.* doi:10.1038/s41598-019-55781-9

2017 Kim S-G & Knösche TR. (2017, Oct). On the perceptual subprocess of absolute pitch. *Frontiers in Neuroscience 11: 557.* doi:10.3389/fnins.2017.00557

Kim S-G, Lepsien J, Fritz TH, Mildner T, Mueller K (2017, Jul). Dissonance encoding in human inferior colliculus covaries with individual differences in dislike of dissonant music. *Scientific Reports 7: 5726.* doi:10.1038/s41598-017-06105-2

Kim S-G & Knösche TR. (2017, May). Resting state functional connectivity of the ventral auditory pathway in musicians with absolute pitch. *Human Brain Mapping 38: 3899–3916*. doi:10.1002/hbm.23637 (final draft: https://bit.ly/2l4dbAh)

2016 Bianco R, Novembre G, Keller P, Kim S-G, Scharf F, Friederici A, Villringer A, Sammler D. (2016, Aug). Neural networks for harmonic structure in music perception and action. *Neuroimage 142: 454-464*. doi:10.1016/j.neuroimage.2016.08.025 (author's proof: https://goo.gl/ApfnbF)

Kim S-G & Knösche TR. (2016, May). Intracortical myelination in musicians with absolute pitch: quantitative morphometry using 7-T MRI. *Human Brain Mapping 37: 3486–3501*. doi:10.1002/hbm.23254

2011 Kim S-G, Kim JS, Chung CK. (2011, Feb). The effect of conditional probability of chord progression on brain response: an MEG study. *PLoS ONE* 6(2*): e17337*. doi:10.1371/journal.pone.0017337

*Journal papers on translational neuroimaging (N = 9)*

2021 Weidacker K, Kim S-G, Nord CL, Rua C, Rodgers CT, Voon V. (2021, June). Avoiding monetary loss: a human habenula functional MRI ultra-high field study. *Cortex*. doi:10.1016/j.cortex.2021.05.013

Zhang C*, Kim S-G*, Li J, Zhang Y, Lv Q, Zeljic K, Gong H, Zhan S, Jin H, Sun B, Wang Z, Voon V. (2021, Jan). Anterior limb of the internal capsule tractography: relationship with capsulotomy outcomes in obsessive-compulsive disorder. *Journal of Neurology, Neurosurgery, and Psychiatry*. doi:10.1136/ jnnp-2020-323062 (*Both share first authorship)

2020 Weidacker K*, Kim S-G*, Callesen MB, Thomsen KR, Voon V. (2020, Nov). Insular and subcallosal cingulate myeloarchitecture and prediction of resilience of alcohol drinking behaviours in youth. *Psychological Medicine.* doi:10.1017/S0033291720003852 (*Both share first authorship)

2019 Zhang C, Kim S-G, Zhang YY, Li Y, Yan FH, Voon V, Sun BM. (2019, May). Habenula deep brain stimulation for refractory bipolar disorder. *Brain Stimulation*. doi:10.1016/j.brs.2019.05.010

Mandali A, Weidacker K, Kim S-G, Voon V. (2019, May). The ease and sureness of a decision: evidence accumulation of conflict and uncertainty. *Brain 142 (5): 1471–1482.* doi.org/10.1093/brain/awz013

Nord C*, Kim S-G*, Thomsen KR, Callesen MB, Kvamme TL, Jensen M, Pedersen MU, Voon V. (2019, Feb). The myeloarchitecture of impulsivity: premature responding in youth is associated with decreased myelination of ventral putamen. *Neuropsychopharmacology 44: 1216–1223*. doi: 10.1038/s41386-019-0343-6 (*Both share first authorship)

2017 Yoon YB, Shin W-G, Lee TY, Hur J-W, Cho KIK, Sohn WS, Kim S-G, Lee K-H, & Kwon JS. (2017, May). Brain structural networks associated with intelligence and visuomotor ability. *Scientific Reports 7: 2177*. doi:10.1038/s41598-017-02304-z

2015 Kim S-G, Kim SN, Jung WH, Jang JH, Kwon JS. (2015, Jun). Alterations of structural networks in patients with obsessive-compulsive disorder: A multimodal analysis of MRI and DTI using mCCA+jICA. *PLoS ONE 10(6): e0127118.* doi:10.1371/journal.pone.0127118

2013 Kim S-G, Jung WH, Kim SN, Jang JH, Kwon JS. (2013, Jul). Disparity between dorsal and ventral networks in patients with obsessive-compulsive disorder: Evidence

revealed by graph theoretical analysis based on cortical thickness from MRI. *Frontiers in Human Neuroscience 7:302*. doi:10.3389/fnhum.2013.00302

*Peer-reviewed conference papers on neuroimaging methods (N = 6)*

2014 Kim S-G[+], Stelzer J, Bazin P-L, Viehweger A, Knösche TR. (2014, May). Group-wise analysis on myelination profiles of cerebral cortex using the second eigenvector of Laplace-Beltrami operator. In *proceedings of the 11th IEEE International Symposium on Biomedical Imaging (ISBI), pp.1007-1010* (selected for [+]oral presentation). doi:10.1109/ISBI.2014.6868043

Chung MK, Kim S-G, Schaefer SM, van Reekum CM, Peschke-Scmitz L, Sutterer M, Davidson RJ. (2014, March) Improved statistical power with a sparse shape model in determining aging effect in hippocampus and amygdala. In *proceedings of SPIE 9034, Medical Imaging 2014: Image Processing, 90340Y.* doi:10.1117/12.2036497

2012 Kim S-G, Lee, H., Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. (2012, May). Agreement between the white matter connectivity based on the tensor-based morphometry and the volumetric white matter parcellations based on diffusion tensor imaging. In *proceedings of the 9th IEEE International Symposium on Biomedical Imaging (ISBI), pp. 42-45* (acceptance rate: 42%). doi:10.1109/ISBI.2012.6235479

Kim S-G, Chung MK[+], Schaefer SM, van Reekum CM, Davidson RJ. (2012, Jan). Sparse shape representation using the Laplace-Beltrami eigenfunctions and its application to modeling subcortical structures. In *proceedings of IEEE workshop on Mathematical Methods in Biomedical Image Analysis (MMBIA), pp. 25-32* (selected for [+]oral presentation; selection rate: 22%). doi:10.1109/MMBIA.2012.6164736

2011 Kim S-G[+], Chung MK, Seo SH, Schaefer SM, van Reekum CM, Davidson RJ. (2011, Nov). Heat kernel smoothing via Laplace-Beltrami eigenfunctions and its application to subcortical structure modeling. In *Pacific-Rim Symposium on Image and Video Technology (PSIVT), Part 1, Lecture Notes in Computer Science (LNCS), 7087: pp. 36-57* (acceptance rate: 42%; selected for [+]oral presentation). doi:10.1007/978-3-642-25367-6_4

Kim S-G, Chung MK, Hanson JL, Avants BB, Gee JC, Davidson RJ, Pollak SD. (2011, Mar). Structural connectivity via the tensor-based morphometry. In *proceedings of the 8th IEEE International Symposium on Biomedical Imaging (ISBI), pp. 808-811.* (acceptance rate: 46%). doi:10.1109/ISBI.2011.5872528

*Books (N = 1)*

2017 Kim S-G. (2017). Myeloarchitecture and Intrinsic Functional Connectivity of Auditory Cortex in Musicians with Absolute Pitch. (Dissertation). In MPI series in human cognitive and brain sciences: 184

*Open-access software (N = 3)*

2014 ANTs parallelization on Condor. https://github.com/solleo/ANTs-Condor

2018–2022 Automatic processing using SPM12. https://github.com/solleo/myspm

2021–2022 Cortical surface visualization. https://github.com/solleo/surfviz

*Selected conference contribution (N = 4)*

2023 Kim S-G, Overath T, Sammler D. (2023, August) Emotion-relevant Representations of Music Extracted by Convolutional Neural Networks Are Encoded in Medial

Prefrontal Cortex. *The Joint Conference of the 17th International Conference on Music Perception and Cognition (ICMPC) and the 7th Conference of the Asia-Pacific Society for the Cognitive Sciences of Music (APSCOM), Tokyo, Japan*. doi:10.6084/m9.figshare.24085104

2020 Kim S-G, de Martino F, Overath T. (2020, Oct). Neural encoding of phonemes modulated by linguistic information. *Neuromatch Conference, Virtual*. https://youtu.be/6hmtrhlhbOc

2014 Kim S-G, Stelzer J, Schulze K, Viehweger A, Knösche TR. (2014, May). Local morphology of auditory cortex in musicians with absolute pitch. *The 5th conference for Neuroscience and Music, Dijon, France.* (Selected for Conference Scholarship)

2011 Kim S-G, Chung MK, Jung WH, Jang JW, Kwon JS. (2011, Nov). Application of network analysis based on cortical thickness to obsessive-compulsive disorder patients. *The Biannual Meeting of South Korean Society of Human Brain Mapping (KHBM), Seoul, South Korea.* (Excellent Oral Award)

## Academic Distinctions

2023–2026 Academy Fellowship, Johanna Quandt Young Academy, Goethe University Frankfurt, Frankfurt am Main, Germany.

2021 Postdoc Research Grant Award, The Charles Lafitte Foundation Program for Research in Psychology & Neuroscience, Duke University, NC, USA.

2017 Summa cum laude, PhD examination, Institute of Psychology. University of Leipzig, Germany.

2014 Conference Scholarship, The 5th Conference for Neurosciences and Music, Mariani Foundation, Italy.

2012–2016 Ph.D. Scholarship (stipend), International Max Planck Research School (IMPRS) on Neuroscience of Communication, Germany.

2011 Excellent Oral Presentation Award, The Biannual Meeting of Korean Society of Human Brain Mapping, South Korea.

2010 Best Poster Presentation Award, The Biannual Meeting of Korean Society of Human Brain Mapping, South Korea.

2001–2004 B.A. Scholarship (full tuition), Suam scholarship foundation, South Korea.

## Statement of Declaration

I hereby declare that all the information provided above is correct and accurate to the best of my knowledge.

Frankfurt am Main, 24 March, 2024

Seung-Goo Kim